Messaging

Text messaging involves sending and receiving electronic messages via mobile networks. Most texts are exchanged using the Short Message Service (SMS), which permits the sending of short, text-only messages of up to 160 characters. However, the Multimedia Messaging Service (MMS) uses mobile networks to exchange messages containing photographs, videos, and audio clips.

How a text is sent

The sender's text is transmitted, via a cell tower, to a mobile switching center (MSC), which finds the address of the sender's short message service center (SMSC) and relays the text there. The SMSC checks whether the recipient is available. If so, it delivers the text via an MSC. If not, the text is stored until the recipient is available.



EVERY **SMARTPHONE** CONTAINS PRECIOUS METALS INCLUDING **GOLD, SILVER**, AND PLATINUM



Internet
Smartphones can connect to the Internet through Wi-Fi or cellular networks. Most phones now use 4G—the fourth generation of mobile technology, which allows much quicker loading speeds.

Games station
Smartphones can be used as portable game consoles. Unlike consoles, they do not include dedicated graphics cards; instead, they feature a powerful graphics processing unit, which renders images, animation, and video.

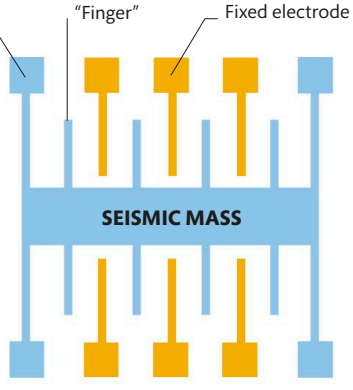
Address book
Most smartphones feature electronic address books where contact information is recorded. Some can pull in information via social media sites and email accounts and can be accessed via voice commands to digital assistants.

Payment systems
Smartphones can make contactless payments via various methods, including radio signals and magnetic signals imitating magnetic strips on bank cards. Payment usually requires an authentication process to confirm identity.

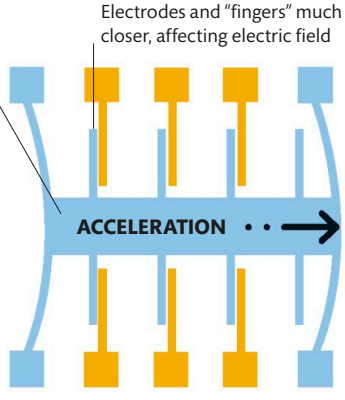
Music
Music can be downloaded from apps, streamed via Wi-Fi or a cellular connection, or imported from a user's collection. Smartphones support many music file formats, including MP3, AAC, WMA, and WAV.

Accelerometers

Many smartphones have miniaturized accelerometers, which measure acceleration. These sensors are used to detect the orientation of the device, so the display can change between landscape and portrait mode depending on how the device is held. They can also be used as an input for pedometers and mobile games.



1 Phone not moving
Fixed electrodes sit between the "fingers" of a comblike seismic mass made of silicon. The electrodes and mass are connected to the battery, so the mass is electrically charged, creating an electric field between the "fingers." When the mass is still, no current flows.



2 Movement detected
The mass deflects in response to motion, and its electric charge affects the electric field around the electrodes, creating a current. This information tells the processor how much the phone is moving and in which direction.